

RayGlides 3W/4W EV Kit Cooling Controller - Complete Schematic

This document describes the complete schematic design, net names, and connectivity details for the RayGlides Cooling Controller PCB.

— — —

1. ASCII Circuit Schematics

A. Input Filter and Overvoltage Protection Stage

[illegible]

B. 12V Synchronous Buck Stage (LM5017)

```

L2: 220uH
Raw_EV_48V ?????? [ LM5017 ] ?????????????????????? 12V
                ?      SW (Pin 8)      ?
                [C3: 1uF]                ???[D3: Boot]    ???[C5: 10uF 25V]
100V Cer.                ? (1N4148)      ?
                ?      [C4: 0.01uF]      ???[R1: 88.7k]
                ?      ?                  ?
GND_PWR ?????????????????????????????????????????????????????????????
                ?
                ????????????????? GND PWR

```

C. Fan Switching MOSFET Stage

```

+12V
?
[ J2: Fan Connector ]
Pin 1 Pin 2
? ?
? [ D4: Flyback Diode 1N4148 ]???
? ?
? ?????????????????????????????????
? ?
? ????
? ?
? [ Drain ]
? MOSFET Q1 (A03400A)
? [ Gate ] [ Source ]
? ? ?
ESP32_GPI018 ??[ R4: 100R ]? ?
? ?
[ R5: 10k ] ?
? ?

```

GND_PWR	??	GND_PWR
---------	--	---------

D. Analog Temperature Sensor Interface (LM35)

+5V	??
	?
	[C8: 100nF]
	?
	?
GND_SENS	??
	?
	[R7: 200R]
	(Pulldown)
	?
GND_SENS	??
	?
	[LM35]
	VCC V_OUT (Pin 2)
	?
	?
	????[R6: 1k]??????? ESP32_GPIO4 (ADC1_CH3)
	?
	[C9: 100nF]
	[D5: TVS 3.3V]
	GND_SENS

2. Component Pinout and Interconnect Netlist

Net Name	From Component (Pin)	To Component (Pin)	Description
:--- :--- :--- :---			
48V_IN	J1 Vehicle Power (1)	F1 Fuse Input (1)	Raw vehicle DC positive line
GND_VEH	J1 Vehicle Power (2)	D2 TVS Anode (2)	Ground return line from vehicle battery
Raw_EV_48V	F1 Fuse Output (2)	D1 Schottky Anode (1)	Internal fused power bus
Protected_48V	D1 Schottky Cathode (2)	L1 Inductor Input (1)	Schottky protected supply line
12V	LM5017 Buck Output (Pin 8 SW)	J2 Fan Output (1)	12V rail for Fan and 5V Buck input
5V	AP63205 Output (Pin 6)	LM35 VCC (Pin 1)	5V rail for Sensor and 3.3V LDO input
3.3V	AP2112K LDO Output (Pin 5)	ESP32-S3 VDD (Pin 3)	3.3V MCU logic supply rail
GND_PWR	LM5017 Ground (Pin 5)	Q1 Source (Pin 3)	Primary buck converter ground plane
GND_SENS	LM35 Ground (Pin 3)	ESP32 Analog Ground (Pin 2)	Isolated quiet analog ground return
ESP32_GPIO18	ESP32-S3 (Pin 22 / GPIO18)	R4 Resistor Input (1)	Fan switching PWM control signal
ESP32_GPIO4	R6 Resistor Output (2)	ESP32-S3 (Pin 8 / GPIO4)	Temperature sensor ADC input line
FAN_GATE	R4 Resistor Output (2)	Q1 Gate (Pin 1)	MOSFET gate driving node
FAN_DRAIN	J2 Fan Pin (2)	Q1 Drain (Pin 2)	MOSFET switching node
TEMP_RAW	LM35 Output (Pin 2)	R6 Resistor Input (1)	Unfiltered analog temperature signal